

George McDonagh

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Experienced Software Engineer with a strong background in real-time systems, performance optimization, and high-efficiency programming. Proven expertise in modern C++ and complex system design, with a deep understanding of low-latency computing. Passionate about building smart, high-performance software.

Senior C++ Engineer (Contract, O/IR35) – Illuvium (04/2025 - present)

Technologies: C++17, Unreal Engine 5 (UE5)

- Responsible for maintaining and extending functionality of the backend real-time simulations in C++17, **using STL and Boost libraries** extensively.
- Optimised **custom network exchange protocols** to minimise message size and maximise throughput.
- Refactored older systems to decrease unnecessary complexity and improve maintainability.

Senior Systems Engineer – Nequinox Studios (09/2023 - 03/2024)

Technologies: C++, Unreal Engine 5 (UE5)

- Performed extensive profiling (**Valgrind, Intel VTune, Wireshark**) which was used to inform **optimizations across multiple real-time systems**, e.g: **client-server** replication. **Average ~25% performance increase.**
- Designed, implemented, and documented off-screen **parallel** rendering pipeline for procedurally generated UI icons.
- Conducted engineering R&D for US-based clients, delivering my findings as written documents and group presentations.

Unreal Engineer (Contract, O/IR35) – Emperia Ltd (07/2022 - 01/2023)

Technologies: C++, Unreal Engine, Python, C#

- Spearheaded the development of an in-house automation tool for processing and rendering 3D environments in **Unreal Engine, C++**.
- Built **Python** middle-ware to handle exchange of content (Finished renders, JSON data) between Unity and Unreal Engine.
- Worked with backend server engineers to design an API for serving our 3D content libraries to user applications.

Lead Programmer (Contract, O/IR35) – Baya Dynamics (01/2022 – 07/2022)

Technologies: C++, UE4/UE5, Node.js

- Redesigned and rebuilt the company's flagship piece of software under serious time constraints, with great success and stakeholder satisfaction. Project completed in 3 of the 5 months expected due to excellent project management and software design.
- Developed from scratch a web application (**Node.js, HTML, CSS**) to connect user display and input to high-fidelity 3D applications running on our servers (**UE Pixel Streaming**).
- Configured and managed high-performance **AWS** servers to serve the application to users.
- Oversaw the hiring process of a small team of Junior developers. Offered mentorship and support throughout my contract.

Programmer – Team 17 Digital Ltd (06/2018 – 05/2021)

Technologies: C++, C#, Unity, Unreal Engine, JavaScript

- Worked on five high-profile game titles, optimizing performance for real-time systems on **Xbox, PlayStation, Windows, and Linux (UE, C++, Unity, C#)**
- Designed and implemented custom low-latency network protocols on top of **UDP** and **TCP/IP** for multiplayer gameplay (**C#**).
- Led build automation and deployment for various platforms, cutting the time between code submission and QA testing in half.

University

BSc in Computer Science & Games Programming – De Montfort University (06/2018) – 2:1 Honours

Technical Skills

- Programming Languages: **C++(17, 20, 23), C#, C, JavaScript, MySQL, Python**

- Networking: **Deep understanding of Client/Server architecture, real-time networking architecture, TCP/IP, UDP socket programming, authentication, RESTful API design**
 - Graphics: **OpenGL, Vulkan, knowledge of modern rendering pipelines**
 - General: **Multithreaded programming**, performance profiling tools (**Valgrind, RenderDoc, VTune, perf**).
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Personal Projects

- High-Performance Rendering Engine – Custom Vulkan/OpenGL engine for real-time 3D rendering (C++)
- TCP/IP chat server – C99, POSIX sockets, Winsock2, custom message protocol
- Sokoban Puzzle Game – Built in Dart/Flutter (Android) and JavaScript (Web).
- Full-Stack personal website – <https://georgemcdonagh.dev> (HTML, CSS, JavaScript, PHP, self-hosted)